MIL-P-46322B(MU)
26 January 1970
SUPERSEDING
MIL-P-46322A(Mu)
17 July 1964

MILITARY SPECIFICATION

PERISCOPE TANK: M42

1. SCOPE

1.1 This specification covers me type of periscope known as Periscope Tank: M42.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein .

SPECIFICATIONS

Federal

MMM-A-131	Adhesives, Glass to Metal.- for Bonding. . . . . of Optical Elements

Military

MIL-G-174	Glass, Optical
MIL-C-675	Coating of Glass Optical Elements
MIL-M-13508	Mirrors, Glass, Front Surface Aluminized, for Optical Elements
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, packaging and Packing of
MIL-I-45607	inspection Equipment, Supply and Maintenance of
MIL-B-60621	Bonding Optical Assemblies (Glass to Metal)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:33Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Army, Frankford Arsenal

F7560085	Fixture, Vibration
D7560124	Milli-Dioptometer 20X
F7645543	Periscope Tank: M42
F8565556	Pressure Tester-Two Station

PACKAGING DATA SHEET

7645543	Packaging of Periscope Tank: M42

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The periscope shall be manufactured in accordance with Drawing and drawings pertaining thereto.

3.2 General specification- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.3 Environmental.

3.3.1 Storage temperatures.- The periscope shall show no evidence of damage or physical failure and shall meet the requirements of this specification at a standard ambient temperature of between +60° and +90° Fahrenheit (F) after having been exposed to, and thermally stabilized at, -80° and +160°F.

3.3.2 Vibration.- The periscope shall be vibrated at a frequency of 30 cycles per and an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ±15 seconds. Subsequent to vibration there shall be no evidence of damage or physical failure and the periscope shall meet the following requirements of this specification.

3.4 Cleanliness.- The periscope shall meet the cleanliness require-, ments of MIL-C-13830 and, in addition, the periscope shall not contain more than 6 particles of dirt. No particle shall be in excess of the area equivalent to a size 50 dig.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:33Z
Check the source to verify that this is the current version before use.

3.5 Sealing.- The periscope shall show no evidence of leakage when subjected to an internal pressure of dry nitrogen at 5 pounds per square inch gage (psig) for a period of two hours. Subsequent to meeting this requirement, the interior atmosphere of the periscope shall contain nitrogen with a dew point of at least as low as -25°F and the pressure shall be 0.5 to 1.0 psig.

3.6 Performance.

3.6.1 Collimation.- With the incident line of sight perpendicular to the mounting surface, light deviation of the emergent line of sight shall not exceed two minutes of arc in any meridian.

3.6.2 Spherical power and astigmatism.- The spherical power and astigmatism of the periscope shall not exceed five millidiopters when using a telescope of at least two inch aperture.

3.6.3 Image tilt.- The image of a plumb line shall be vertical within one degree.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein." Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-O-13830 and MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three periscopes. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the-contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:33Z
Check the source to verify that this is the current version before use.

4.2.2 contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the eontract. The sample and a record of this inspection including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 components and subassemblies.- All components and subassemblies shall be inspected In accordance with the inspection provisions contained herein using approved methods, procedures and equipment to determine conformance to component drawing requirements. In the absence of specific test methods and procedures, the methods and procedures in MIL-O-13830 and MIL-F-13926 shall be utilized as applicable. Certified test results, identifiable with the examination, may be accepted in lieu of repeated tests if such tests will be destructive to the end item or its components. The Government reserves the right to require retests when deemed necessary to assure acceptable components.

4.3.2.1.1 Certification.- Certification shall Include test data and results for all characteristics specified in Table III. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval, compliance with the certification provisions of this specification and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS CHaracteristic	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65 percent defective		
101. Vibration	3.3.2	4.6.2, 4.6.2.1
102. Cleanliness	3.4	4.6.3
103. Collimation	3.6.1	4.6.5, 4.6.5.1

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T05:33Z
Check the source to verify that this is the currentversion before use.

TABLE I - Cont'd

104.	Spherical power and astigmatism	3.6.2	4.6.5, 4.6.5.2
105.	Image tilt	3.6.3	4.6.5, 4.6.5.3

MINOR: NONE

NOTE: The tests in Table I shall be conducted in accordance with the test condition requirements in the "Test Facilities" requirement of MIL-F-13926.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Control tests.

4.4.1 Special sampling.- Samples shall be selected from lots at random using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table 1 prior to and after being subjected to the examinations and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Storage temperature	3.3.1	4.6.1
302.	Sealing	3.5	4.6.4

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.4.3 General.- Unless otherwise specified, certification shall be provided for the characteristics in Table III. Certification shall be in accordance with the provisions in 4.3.2.1 and 4.3.2.1.1.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST DATA TO COMPLY WITH
303.	Fabrication (general)	3.1	Applicable dwgs (see 3.1) MIL-O-13830 & MIL-F-13926.
	Fabrication (components)	3.1	Applicable dwgs (see 3.1); Inspection dwgs C7641866, D7680600 & D7680606; and Specification;
304.	Optical glass		MIL-G-174
305.	Optical element bonding		MIL-B-60621
306.	Optical element performance		MIL-O-13830, 4.6.5.2
307.	Optical element surface quality		MIL-O-13830
308.	Optical element coating		MIL-C-675

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:33Z
Check the source to verify that this is the current version before use.

TABLE III - Cont'd

309.	Optical element aluminized surface		MIL-M-13508
310.	Optical element polished surface		MIL-O-13830
311.	Optical element adhesive		MMM-A-131
312.	General specification ("a" through "e")	3.2	Applicable dwgs (see 3.1) and MIL-F-13926

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor "shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL) when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is Intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.- Special test equipment in Table IV, comparable to referenced drawings and having at least the accuracy depicted thereon, shall be provided. It shall be utilized to conduct periscope and component tests related to the examination listed herein.

TABLE IV

TEST EQUIPMENT	DRAWING	EXAMINATION
Fixture, Pressure Tester - Two Station	F8565556	Periscope (only)
Fixture, Vibration	F7560085	Periscope (only)
Adapter, Vibration Fixture	F7645543 & F7560085	Periscope (only)
Milli-dioptometer	D7560124	Periscope and components

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:33Z
Check the source to verify that this is the current version before use.

4.6 Test methods and procedures.

4.6.1 Storage temperatures.- The periscope shall be subjected to each temperature extreme in 3.3.1 until its exposed surface is thermally stabilized at each extreme temperature. Upon completion of both exposures, the periscope shall be allowed to thermally stabilize in an ambient temperature range of+60°F to+90°F. The periscope shall then be examined for evidence of physical damage and subjected to the following tests to determine conformance to 3.3.1.

4.6.2 Vibration.- Standard vibrating equipment capable of providing the amplitude, frequency, and time duration specified in 3.3.2 shall be utilized for the vibration test. The periscope shall be vibrated in a vertical plane. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926.

4.6.2.1 Post vibration.- Subsequent to vibration and prior to conducting the remaining tests, the periscope shall be given a visual and tactile examination for physical defects resulting from poor workmanship or materials. In addition to the visual and tactile examination, the periscope shall be vigorously shaken by hand to detect any audible evidence of internal defects. Periscopes which exhibit no evidence of damage shall be subjected to the following tests to assure compliance with 3.3.2.

4.6.3 Cleanliness.- The optical surfaces of the periscope shall be examined for cleanliness in accordance with MIL-O-13830 utilizing the procedure herein to determine conformance to 3.4. Employing transmitted light, hold the periscope with the light source at the upper window and conduct the examination while viewing through the lower window. Repeat the procedure by illuminating the lower window while viewing through the upper window.

4.6.4 Sealing.- Utilize test equipment specified in 4.5.2.3 to conduct this test. Pressurize the periscope with dry nitrogen gas to the specified value and observe no loss in pressure for a minimum period of two hours. Subsequent to this test, dry nitrogen gas shall be used with a commercially available dew point tester to flush the periscope until the dew point of the bled gas and its internal pressure is as specified in 3.5. The periscope shall then be again sealed. This test shall be performed subsequent to test 4.6.1 and 4.6.2.

4.6.5 Performance.- Tests for collimation, spherical power and astigmatism, and image tilt shall utilize a calibrated target, auxiliary telescope (with parallax removed) and a holding medium to support and position the periscope under test. The holding medium shall assure that the entrance ray is perpendicular to the periscope mounting surface and also that the offset distance - between intersection of the auxiliary telescope's reticle lines and intersection of the target lines - equals the distance between the geometric centers of periscope windows 7645551 and 7645552. The test target shall have a horizontal and vertical cross-line with limit marks in each meridian to measure deviation in the center of the "field of view (FOV). If a wall target is used, the dimension of one mil shall be computed by the formula: Mil Size = $\frac{\text{Target Distance (in inches)}}{1000}$.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:33Z
Check the source to verify that this is the current version before use.

Prior to conducting periscope tests the telescope vertical line and the target vertical line shall be made plumb, and the holding medium made level on a qualified surface.

4,6.5.1 Collimation.- This test shall be performed under the conditions of 4.6.5 and in accordance. with 3.6.1. To determine conformance to 3.6.1, direct a line of sight through the telescope and periscope to the target, and observe in each median compliance with 3.6.1.

4.6.5.2 Spherical power and astigmatism.- Utilize a four line resolution chart as depicted in MIL-O-13830 and a milli-dioptometer (see 4.5.2.3). Employ the setup and operating instructions on Drawing D7560124 as applicable to determine conformance to 3.6.2 in any two target line sets which are 90° with respect to each other.

4.6.5.3 Image tilt.- Utilize an auxiliary telescope with a vertical reticle line having a FOV which subtends an angle greater than 80 mils and a calibrated target. The target shall have a vertical line and appropriate markings-to indicate - over an 80 mil FOV - a displacement of the periscope's line of sight from its vertical line which subtends a 1.5 mil angle. To determine conformance with 3.6.3 establish coincidence between the vertical reticle line of the auxiliary telescope and the vertical target line image - while sighting through the periscope - at one extremity of the 80 mi1 FOV. Observe across the 80 mil FOV no displacement of the targets vertical line image that will subtend an angle greater than 1.5 mil,

5. PREPARATION FOR DELIVERY

5.1. Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 7645543. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Periscope Tank: M42 is a unity power periscope primarly intended for use with Elbow Telescope: M118C on the 155 MM Self-Propelled Howitzer: M109. The periscope offsets the line of sight of the telescope to clear the visual obstruction created by the blast deflector on the howitzer gun tube.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:33Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A048

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:33Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES F[illegible] MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202640-22T05:33Z
Check the source to verify that this is the current version before use.